SYNTHAGENT: Reliable LLM-Based Synthetic Data Generation for Small Healthcare Datasets

Sancho Amaral Simões[1,*], João P. Vilela[1,2], Pedro Henriques Abreu[1]
[1]*University of Coimbra, CISUC/LASI - Centre for Informatics and Systems of the University of Coimbra, Department of Informatics Engineering*, Coimbra, Portugal
[2]*CRACS/INESCTEC-DCS - FCUP, University of Porto*, Porto, Portugal
[*]Corresponding author
sanchosimoes@student.dei.uc.pt, jvilela@fc.up.pt, pha@dei.uc.pt

***Abstract*—Classification tasks in healthcare, such as diagnosing a disease or anticipating an adverse outcome, often have to be learned from small datasets, because the disease is rare or ~~because~~ labelling records is expensive~~, and synthetic~~. Synthetic data generation can improve ~~the performance of~~ such classifiers by adding artificial records to the training data. ~~The generatorsin the literature~~Existing generators, however, do not address the difficulties of small data: oversampling techniques and deep generative models learn only from the few available records, recombining patterns that already exist in them, and large language models (LLMs), which add domain knowledge from pre-training, return text that often contains fewer records than requested, malformed records or values of the wrong type. ~~In this context, we~~ We developed SYNTHAGENT, an LLM-based synthetic data generation system for small healthcare datasets whose control layer infers the columns and types of the real data, checks every ~~response of the LLM~~ LLM response, asks for ~~the correction~~ corrections of invalid values and requests new records until the configured number is reached, ~~while~~ recording the cost of every request. SYNTHAGENT was tested on ten public healthcare datasets with seven LLMs from two providers and compared with three LLM-based and ~~eight~~ ten established generators. It was the only LLM-based generator ~~that delivered every synthetic dataset at the configured size, at a small additional cost in tokens, and, with the strongest LLM, it gave~~ with no integrity violations: no missing, malformed or mistyped records, so its output can enter a training pipeline without inspection, while needing fewer tokens per delivered record than direct prompting. Across the ten datasets, its records reproduced the real distributions reasonably well, kept a membership-inference attack near chance on every dataset and were near-copies of real records less often than those of oversampling, and SYNTHAGENTachieved the largest median F1-score gain among the ~~fourteen~~ sixteen generators.**

***Index Terms*—small data, synthetic data generation, large language models, healthcare**

I. INTRODUCTION

Many prediction tasks in healthcare, such as diagnosing a disease or anticipating an adverse outcome, ~~suffer from a lack of~~ have too little data. A rare disease may have only a few dozen diagnosed cases, a ~~study conducted in a single healthcare facility~~ single-facility study is limited to the patients it can recruit, and labelling records often requires expert review. ~~Data sharing~~ Sharing data between facilities could ~~increase the number of~~ add records, but ~~it is frequently restricted by privacy regulations~~ privacy regulations often restrict it [1], so each facility often has to learn from its own data. ~~In our work we refer to such datasets as~~ We call such datasets *small data*: datasets whose number of records is too low to train a reliable prediction model, a limit that depends on the number of features, the number of classes and the frequency of the outcome [2]. A common ~~way to mitigate~~ remedy for small data, and for a lack of data ~~more generally~~in general, is synthetic data generation (SDG), which adds to the training set artificial records that follow the distribution of the real ones~~to the training set~~ [3]. This paper focuses on SDG for prediction tasks on small healthcare datasets.

SDG is, however, hardest when data are small. Oversampling techniques, such as the Synthetic Minority Oversampling Technique (SMOTE) [4], and deep generative models, such as the conditional tabular generative adversarial network (CTGAN) and the tabular variational autoencoder (TVAE) [5], learn the distribution from the same few records they are meant to complement, recombining patterns that already exist in those records, for example by interpolating between two existing patients [6, Sec. 2.8]. Large language models (LLMs) offer ~~a new perspective to address such issue~~another route. They are pre-trained on large text corpora, which gives them domain knowledge about healthcare variables, such as plausible ranges and typical relations between them, and they can generate new records from a few real examples placed in the prompt, without any training on the target dataset [7], [8]. Their output, however, is text that has to be parsed into records, and nothing guarantees that it forms a dataset: a response may contain fewer records than requested, malformed records, missing or invented columns, or values of the wrong type (Figure 1). ~~From a data-management perspective~~For data management, an LLM is thus an unreliable data source: like any automatic ~~producer of data~~ data producer, its output should satisfy integrity constraints before it is loaded into a training set ~~[9], [10]. Every augmentation method is configured with~~ [9]–[12]. Two integrity constraints apply: the dataset must have the configured number of records, and every record must have the columns of the real data with values of the right type. First, the number of synthetic records ~~to add~~is a design decision, usually a proportion of the real training set ~~[4], and~~ chosen, for example, to balance

the classes [4]; a generator that ~~silently returns fewer or malformed records~~ returns 64% of them leaves the minority class under-represented, which is the problem augmentation was meant to solve. Second, synthetic records are typically loaded into a training pipeline without inspection, where a missing column or a value such as `age="sixty"` either stops the job or is silently turned into a missing value. Moreover, datasets of different sizes cannot be compared fairly across generators or runs, nor exchanged between facilities in an agreed format. A generator that violates these constraints thus corrupts the training data without the user noticing. In addition, each request to a commercial LLM has a monetary cost, proportional to the ~~number of~~ tokens (word fragments) processed, which is rarely reported.

~~One problem that stands out~~ The problem is therefore how to obtain from an LLM a synthetic dataset with the configured number of records, the columns and types of the real data and a known cost ~~, so that it can be used to improve the~~ that can improve classifiers trained on small data. Formally, given the training records D of a small dataset, their columns and types S and the configured number of synthetic records N, a configuration θ of the generator produces a synthetic dataset G_θ. The goal is to find a configuration whose dataset belongs to the set $\mathcal{F}(S, N)$ of datasets with exactly N records that conform to S and that maximizes the gain in classifier performance ΔU, while keeping the distance from the real distribution L_{fid}, the privacy risk L_{priv} and the cost κ low:

$$\theta^\star \in \underset{\theta:\, G_\theta \in \mathcal{F}(S,N)}{\arg\min} \big(-\Delta U,\ L_{\text{fid}},\ L_{\text{priv}},\ \kappa\big)(\theta). \qquad (1)$$

~~The constraint follows the data-quality dimensions of completeness and conformity~~ $\mathcal{F}(S, N)$ encodes two integrity constraints, the requested number of records and the columns and types of the real data [13], [14], and the objectives are the utility, fidelity and privacy on which synthetic data are evaluated [3], [15], together with cost: ΔU is the change in classifier performance when G_θ is added to D, L_{fid} the divergence between the real and synthetic distributions, L_{priv} the success of a membership-inference attack, and κ the number of tokens processed; Section V defines how each is measured. The objectives are kept separate rather than combined with weights, because the right trade-off depends on ~~the use of the data~~ how the data are used.

~~Currently, deep~~ Deep generators [5], [16], [17] and language models fine-tuned on the target data [7], [18] must be trained on the scarce records ~~, and they~~ and were designed and evaluated on datasets with tens of thousands of records. Approaches that prompt an LLM directly, such as CLLM [8] and EPIC [19], avoid training, but they focus on ~~the design of the prompt and on the selection of~~ prompt design and on selecting the generated records; they do not ensure that the output has the configured number of records and the correct columns and types, and they do not report its cost. Constrained decoding, now offered by LLM providers, fixes the format of each response but not the ~~size of the dataset~~number of records. Finally, generators are usually compared on ~~the result of a~~ single run, which is unreliable in small data. ~~A small test set amplifies chance: with~~ With 40 test records, one misclassified record changes accuracy by 2.5 percentage points, ~~and the randomness of the data split and of the LLM's sampling makes~~ so two runs of the same generator ~~differ by amounts comparable to the differences being compared~~can differ as much as two generators.

~~To address these limitations, we~~ We propose SYNTHAGENT (*synthetic data agent*), an SDG system for small healthcare datasets that combines a pre-trained LLM, which proposes new records, with a *control layer*, a deterministic ~~software~~ component that checks every ~~response of the LLM~~ LLM response before any of its records ~~is accepted into~~ enters the synthetic dataset. SYNTHAGENT treats generation as a data-ingestion problem: the control layer selects the real records shown to the model, repairs and validates every response against the columns and types of the real data and asks for corrections, requests exactly the missing records, and records the provenance and cost of every request in an audit trail. Like an LLM agent, ~~the system thus alternates actions , the~~ it alternates actions (requests to the LLM~~, with observations , the~~) with observations (checks of their output~~, which gives the system~~), hence its name. ~~This point is crucial since completeness and conformity~~ The control layer is key: the integrity constraints then hold by construction rather than by trust in the LLM, and ~~since~~ each healthcare facility can run the system on its own data and obtain a dataset that can be audited like any other data source.

~~The main contributions of this paper~~ Our contributions are:

- SYNTHAGENT, an SDG system for small healthcare datasets that requires no training on the target data and whose control layer enforces ~~completeness and conformity~~ integrity by construction: it delivers the configured number of records with the columns and types of the real data, ~~together~~ with an audit trail of every request and its cost;
- a deployment pattern in which each healthcare facility runs SYNTHAGENT locally and only model updates are shared, so that no record leaves the facility;
- an open-source implementation that works with OpenAI-compatible and Anthropic LLM endpoints, including locally served models.

~~Our objective is to determine~~ We ask whether SYNTHAGENT makes LLM-based SDG ~~a reliable and useful option~~ deliver datasets of assured integrity that are useful for small healthcare ~~datasets. To this end, four main research questionsemerged in this work~~data, through four research questions:

RQ1 ~~Is there any relationship between~~ Does the control layer ~~and the reliability~~ ensure the integrity of the generated datasets?

RQ2 Do the generated records preserve the distribution of the real data without reproducing the real records?

RQ3 Do the generated records improve the performance of classifiers trained on small data?

RQ4 How does SYNTHAGENT compare with established generation strategies, and how much do its results depend on the LLM?

~~To answer these questions, an experimental setup was built with~~ We answer them on ten public healthcare datasets ~~containing between~~ with 116 ~~and~~ to 768 records. For each dataset, synthetic records were produced by SYNTHAGENT with commercial LLMs from two providers, by three LLM-based baselines (direct prompting, with and without the provider's option to force responses into a fixed format, and the curation approach of CLLM), and by ~~eight~~ ten established generators from the oversampling, statistical, tree-based and deep families. Each dataset was randomly divided ten times into a training and a test set, and on each division ten classifiers were trained with and without synthetic records and compared on the same test set. The main conclusions ~~are the following~~follow. Without the control layer, the LLM never returned the configured number of records, even when its responses were forced into the format of the data, whereas SYNTHAGENT ~~completed every dataset at only 1.9% more cost~~ always did, with 17% fewer tokens per delivered record (RQ1). The synthetic records reproduced the distribution of each real feature with a fidelity of 72 to 77 out of 100, and a membership-inference attack could not reliably identify the real records used, although a classifier could still tell synthetic from real records (RQ2). The effect on classifiers depended on the dataset, the classifier and the LLM: ~~with the strongest LLM,~~ the median F1-score gain ~~was~~ reached 1.4 percentage points (RQ3~~). With that LLM,~~ , RQ4). SYNTHAGENT gave the largest median gain among ~~fourteen~~ sixteen generators; a curation filter improved F1 more often but by less and ~~delivered incomplete datasets, oversampling varied the most,~~ returned fewer records than configured, and tree-based generators reproduced the distributions more closely at a higher privacy risk ~~, and the parameters of the control layer did not change utility beyond the variation between repeated runs~~ (RQ4).

~~The remainder of the paper is organized as follows.~~ Section II ~~presents the backgroundknowledge~~reviews background, Section III describes SYNTHAGENT and Section IV its deployment across healthcare facilities. Section V presents the experimental setup, Section VI reports the results, Section VII discusses ~~their~~ implications and limitations, and Section VIII concludes~~the paper~~.

II. BACKGROUND KNOWLEDGE

~~In this section ,~~ This section reviews the families of ~~methods for synthetic data generation (SDG) of tabular data are first reviewed. Then, the limitations of each family~~ tabular SDG methods, their limitations when data are small~~are discussed, and finally~~ , and the work on validating generated data and ~~on~~ evaluating synthetic data ~~,~~ on which SYNTHAGENT builds~~, is presented~~.

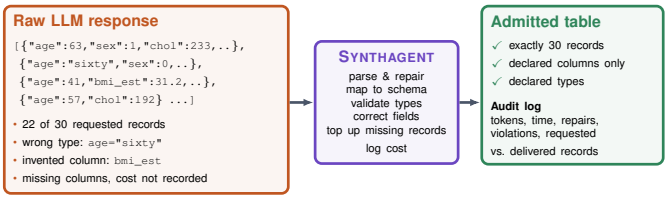


Fig. 1. The problem addressed in this paper, in an illustrative example. Asked for 30 records of a heart-disease dataset, an LLM response (left) can contain too few records, invented or missing features and values of the wrong type. SYNTHAGENT (right) validates and corrects every response, so the resulting dataset has ~~the requested size and schema~~no integrity violations, and a log records each correction and its cost.

A. Synthetic Data Generation for Tabular Data

Tabular SDG methods fall into four families ~~, which~~ that differ in how they produce ~~new~~ records. *Interpolation and statistical methods* derive new records directly from the real ones or from a simple probabilistic model of them: SMOTE places a new minority-class record on the line segment between two neighbouring records of that class [4], and Bayesian networks estimate the dependencies between features and sample new records from them ~~[20]~~ [20], [23]. *Deep generative models* train a neural network to approximate the joint distribution of the features and ~~then~~ sample from it: CTGAN trains a generator against a discriminator, TVAE learns a compressed representation of the records [5], and diffusion models such as TabDDPM, CoDi and TabSyn learn to reverse a gradual corruption of the records [16], [17], [24]; relational and cross-silo variants extend these models to several tables or institutions [25], [26]. *Fine-tuned language models* write each record as a sentence, fine-tune a language model on these sentences and generate new ones, as in GReaT [7], REaLTabFormer [18] and TabLoft [27]. *Prompted LLMs* place a few real records in the prompt of a pre-trained LLM and ask it for new ones, without ~~any~~ training, as in CLLM [8] and EPIC [19].

B. Synthetic Data Generation in Small Data

Most of ~~the previously mentioned~~ these generators were designed and benchmarked on datasets with tens or hundreds of thousands of records, such as Adult (48,842 records) or Covertype (581,012 records) [5], and their behaviour on a few hundred records ~~has received far less attention~~is far less studied. Table I summarizes, for each family, how records are produced and its main limitation when data are small. Interpolation and statistical methods, as well as deep generators, are fitted only to the real records, so they cannot add information beyond what those records contain [6, Sec. 2.8]. They also treat every column as an anonymous number~~: they do not know that a column holds a patient's age or cholesterol level, and therefore cannot use~~ , ignoring what is known about ~~these variables~~ variables such as age or cholesterol. Deep generators and fine-tuned language models have many parameters~~by design~~, which a few hundred records cannot estimate reliably, so they may produce records outside the observed ranges and types, or reproduce training records [28]. Prompted LLMs need no training and bring domain knowledge, but nothing

TABLE I

FAMILIES OF TABULAR SDG METHODS: HOW THEY PRODUCE RECORDS, THEIR MAIN LIMITATION IN SMALL DATA, AND THEIR PROPERTIES. ✓: PROVIDED BY DESIGN; ×: NOT PROVIDED. SMOTE'S NUMBER OF RECORDS IS SET BY THE CLASS BALANCE, NOT BY THE USER; FINE-TUNED LANGUAGE MODELS DISCARD UNREADABLE RECORDS BUT DO NOT CHECK TYPES; CONSTRAINED DECODING IS A VARIANT OF PROMPTED LLMS.

Family	How records are produced	Main limitation in small data	No training on target data	Record count	Columns and types checked	Cost recorded
Interpolation and statistical [4], [20]	Combine real records or sample a simple model fitted to them	Only recombine existing patterns; ignore feature meaning	✓	×	×	×
Deep generative [5], [16], [17]	Train a neural network on the real records and sample from it	Few records to fit many parameters; invalid or copied records	×	✓	×	×
Fine-tuned language models [7], [18]	Fine-tune a language model on records written as text	Training on scarce records	×	✓	×	×
Prompted LLMs [8], [19]	Ask a pre-trained LLM for records, with real examples in the prompt	Unreliable output: ~~wrong size,~~ missing or malformed records, unknown cost	✓	×	×	×
Prompted LLMs, constrained decoding [21], [22]	As above, with the output restricted to a JSON schema	Format guaranteed, but not the number of records; cost unknown	✓	×	✓	×
SYNTHAGENT (ours)	Prompted LLM whose output is checked and completed by a control layer	—	✓	✓	✓	✓

ensures that their output contains the configured number of records with the correct columns and types, and the cost of obtaining it is not reported; CLLM [8] filters the generated records by their learning dynamics and EPIC [19] designs balanced prompts, but both concern the quality of the ~~records that are kept~~ kept records rather than the ~~size, structure~~ integrity and cost of the output. ~~As discussed in Section I, all families are also usually compared on a single run.~~ SYNTHAGENT keeps the advantages of prompted LLMs and adds the control layer that they lack~~, and its evaluation repeats every experiment to measure the variation between runs~~.

C. Validation of Generated Data

The control layer applies to generation a principle long used in data management: data produced by an automatic process are checked against explicit constraints before they are accepted. Kamino enforces integrity constraints during private data synthesis [9], data-cleaning systems combine statistical inference with integrity constraints [29], [30], and recovering structured records from the free-form output of an LLM has been studied as a data-engineering problem [10], as have LLMs for data wrangling, error detection and document processing, which must cope with unreliable model output [31]–[33]. Constrained decoding ~~goes further and~~ restricts the tokens an LLM may emit to ~~those allowed by~~ a grammar or ~~a~~ JSON schema [21], [22], ~~and LLM providers now offer it~~ now offered by providers as a structured-output mode~~. It guarantees~~ ; it fixes the format of each response ~~, but not how many records a response contains, so it does not by itself deliver a dataset of a given size;~~ but not the number of records (Section VI-A~~tests it as a baseline~~). In SYNTHAGENT, the constraints are the columns and types inferred from the real data, and ~~the checks are applied to every response of the LLM~~ every LLM response is checked, in the ~~action–observation~~ action-observation loop of LLM agents [34], [35].

D. Evaluating Synthetic Data

Data-preparation studies have shown that a step that looks beneficial can harm downstream models [36], [37], and evaluation frameworks such as Synthcity and SynthEval recommend assessing fidelity, privacy and utility together [3], [15]. Fidelity can be complemented by a classifier two-sample test, which asks whether a classifier can tell real from synthetic records [38], and privacy by membership-inference attacks, which ask whether an attacker can tell which records were used for generation [28], [39]. Formal guarantees require ~~syntactic anonymization~~ models such as ℓ-diversity~~and~~, t-closeness [40], [41] or differential privacy [42], which SYNTHAGENT does not claim.

III. THE SYNTHAGENT SYSTEM

SYNTHAGENT meets the constraint of Equation (1), a synthetic dataset with exactly N records that conform to S, by construction rather than by trusting the LLM. This guarantee is the core of the system: the LLM only proposes records~~, and~~ ; the control layer decides which ~~of them are accepted~~to accept, asks for corrections or ~~for~~ more records, and stops only when N conforming records have been accepted or a stopping limit is reached. The objectives of Equation (1) are left to the ~~domain knowledge of the LLM and to~~ LLM's domain knowledge and the real records shown to it~~, and~~ ; they are measured, not optimized, in Section VI. The real dataset is never modified.

Figure 2 shows the architecture and Algorithm 1 the procedure. A profiling step first infers the columns and their types S from the training records. Each request then passes through four stages: the context budgeter selects the real records shown to the LLM (1), the LLM proposes a batch of records (2), the deterministic verifier parses, maps and validates it (3), and the admitted records are counted against N (4). The run produces two outputs: the synthetic table G and an audit trail of every request. The rest of this section follows ~~the procedure in this~~

Algorithm 1 Generation procedure of SYNTHAGENT

Require: training records D, columns and types S, number of records N, parameters of Table III

pool $\mathcal{P} \leftarrow$ random sample of $\rho|D|$ records of D
accepted records $A \leftarrow \emptyset$ ▷ empty at the start
repeat
  context $C \leftarrow$ next records of $\mathcal{P}$ that fit in a share ω of the context window W ▷ (1)
  batch $Y \leftarrow$ ask the LLM for $\min(b, N - |A|)$ new records, given S and C ▷ (2)
  records $X \leftarrow$ parse the batch Y, repairing it if needed, and map its columns to S ▷ (3)
  correct the invalid values of X, for at most c rounds ▷ (3)
  $A \leftarrow A \cup X$; log tokens, repairs and violations ▷ (4)
until $|A| \geq N$ **or** $t_{\max}$ rounds **or** $u_{\max}$ empty answers
return synthetic table G, the first N records of A, and the audit trail

~~order ; the parameters it uses,~~ this order and ends with the parameters and the reason for each value~~, are summarized at the end of the section~~.

A. Controller State

At every step t, the control layer keeps an explicit state

$$z_t = (S,\, C_t,\, A_t,\, N - |A_t|,\, B_t,\, E_t), \tag{2}$$

made of the columns and types S, the real records C_t currently shown to the LLM, the synthetic records A_t accepted so far, the number of records still missing $N - |A_t|$, the number of tokens B_t still available in the context window, and the log E_t of parser and validator results. From this state the LLM, denoted π_ϕ, proposes a batch $Y_t \sim \pi_\phi(\cdot \mid z_t)$, and the accepted set grows only through the verifier:

$$A_{t+1} = A_t \cup K\big(V_S\big(P_S(Y_t)\big)\big), \tag{3}$$

where P_S parses the batch Y_t and maps its columns to S, V_S validates the types, and K sends the invalid values back to the LLM for correction. Because the state is explicit rather than implied by the conversation history, the number of missing records is always known, and the parser log E_t records every failure for later inspection.

B. Context Budgeter (1)

The context budgeter decides which real records the LLM sees. It draws a random sample of $\rho|D|$ training records, between 31 and 123 records in our datasets, written as compact JSON with the feature columns and the class, so that the model sees how features and class co-occur. Because a long prompt can exceed the context window W of the LLM, each request shows as many records of the sample as fit in a share ω of the window:

$$\begin{aligned} C_t &= \mathrm{Slice}_t\big(\mathrm{Select}_s(D, \lceil \rho|D| \rceil), k_t\big), \\ k_t &= \max\{k : T(C_{t,k}) \leq \omega W\}, \end{aligned} \tag{4}$$

where Select_s draws the sample with strategy s, T estimates the length in tokens of a prompt with k records, and Slice_t rotates through the sample, so that successive requests see different patients. If the provider still rejects a request as too long, the budgeter shows fewer records and retries. Besides the default random strategy, *dense* prefers records with fewer missing values, and *diverse* selects the records closest to the centres of k-means clusters. Neither strategy is class-stratified, a limitation discussed in Section VII.

C. LLM Proposal and Prompts (2)

The LLM receives the same three-part conversation for every dataset and ~~every~~ LLM (Table II). A first message gives the column names and the sampled records in a compact format, a list of column names followed by one list of values per record. A generation message then asks for an exact number of new records, at most b, and states an output contract: JSON only, the same columns in the same order, a value in every cell, raw numbers, and no additional columns. No free-text description of the dataset or of the prediction task is added, so that no dataset receives more prompt engineering than another. The prompts are therefore deliberately simple, and their wording is not part of the contribution: the guarantees come from the stages that check the response.

D. Deterministic Verifier (3)

The verifier turns a response into records that conform to S, in three steps. *Parsing* extracts the records with a local parser that removes surrounding text and code fences, deletes trailing commas, inserts missing commas between records and closes a response truncated by the output limit. If parsing still fails, the control layer sends one repair request that quotes the position and the text around the error, at a low temperature so that the repair fixes the syntax without changing the content. *Mapping* assigns the returned values to the columns of S: by position when the response has the expected number of columns, and otherwise by matching the returned column names to those of S, ignoring case and spacing; an invented column is dropped and a missing one is left empty. *Validation* checks every value against the type that profiling inferred for its column (integer, real or categorical); optional rules can also bound numerical ranges or restrict categories, but they were off in our experiments except in the range-enforcement variant. The invalid values are grouped by column and sent back to the LLM with one example per column, for at most c correction rounds, so that the model corrects only what is wrong instead of regenerating the batch. The three steps escalate from cheap to costly: a defect that the local parser or the mapping can fix costs no request, a syntax error costs one repair request, and only type violations cost correction rounds. In the example of Figure 1, the parser recovers the 22 records, the mapping drops the invented column `bmi_est` and leaves the missing columns empty, the validator flags `age="sixty"`, and one correction round replaces it with a number; the completion loop then requests the eight missing records. Every step is written to the parser log E_t.

E. Admit and Count (4)

LLMs often return fewer records than requested. The admitted records are counted against N, and after the planned

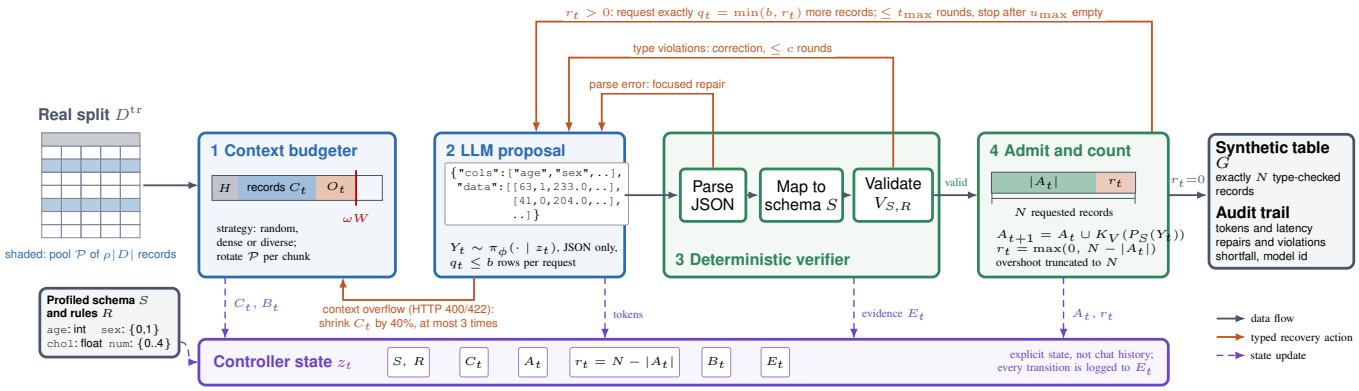


Fig. 2. Architecture of SYNTHAGENT. The context budgeter selects the real records shown to the LLM (1); the LLM proposes a batch of records (2); the deterministic verifier parses it, maps it to the columns and types S and validates it (3); and the admitted records are counted against the configured number N (4). Each failure type triggers its own correction (orange). Every stage reads or updates the controller state z_t (purple). The outputs are the synthetic table G and the audit trail of the run.

TABLE II
WHAT THE LLM RECEIVES IN ONE GENERATION REQUEST. THE PROMPTS ARE THE SAME FOR EVERY DATASET AND LLM. "NOT SUPPLIED" IS INTENTIONAL: THOSE SIGNALS CANNOT INFLUENCE THE GENERATED RECORDS.

Context layer	What the model receives	Why it is included or excluded
Schema	Ordered column names and an exact column-count requirement	Fixes the record interface and prevents invented or omitted fields.
Demonstrations	Compact row-major serialization of a sample of $\rho\|D\|$ training records, with feature and target values	Exposes joint patterns, value formats, label prevalence and plausible cross-column combinations without sending the whole local table.
Generation task	Exact number of new records for the current batch, at most b	Turns an open-ended completion into a bounded action whose output count can be verified.
Output contract	JSON only, fixed column order, complete cells, raw numeric values, no extra metadata	Makes parsing deterministic and removes interface failures unrelated to statistical quality.
Dynamic feedback	Parser error location after malformed JSON, or grouped field violations after failed validation	Reveals only the evidence needed for a targeted repair instead of replaying the full history.
Free-text metadata	Not supplied	Prevents unequal, dataset-specific prompt engineering from confounding the comparison.
Test information	No holdout rows, predictions, metrics or classifier identities	Preserves the evaluation boundary and prevents optimization against the test set.
Global statistics	No separately computed means, quantiles, correlations or class quotas	Fidelity must arise from the selected records; the consequence is weaker protection against rare-mode omission.

batches the completion loop requests only the missing records,

$$r_t = \max(0, N - |A_t|), \qquad q_t = \min(b, r_t), \tag{5}$$

and truncates any excess. It stops when N records have been accepted, after $t_{\max}$ rounds, or after $u_{\max}$ consecutive empty answers; a run that ends short because the provider keeps failing is recorded as such. The run returns the synthetic table G and an audit trail with the tokens processed, the generation time, the repairs, the violations and the requested and delivered numbers of records, which makes a synthetic dataset inspectable after the fact and gives the cost κ in Equation (1) a measured value.

F. Parameters

Table III lists the parameters and the values used in the experiments, fixed beforehand from the size of our datasets and the context and output limits of the LLMs rather than tuned on the results; Section VI-A examines their effect.

IV. DEPLOYMENT ACROSS HEALTHCARE FACILITIES

~~As discussed in Section I, healthcare~~ Healthcare facilities often cannot pool their records (Section I) [1], [43], and SYNTHAGENT was designed for this setting. Because the control layer needs only a local dataset, its schema and an inference endpoint, each ~~healthcare~~ facility of a network can run it inside its own infrastructure. Consider K healthcare facilities, each holding a small dataset D_k for the same prediction task, with different case mix, coding practice and class prevalence. Each ~~healthcare~~ facility runs SYNTHAGENT locally and trains on its real and synthetic records,

$$G_k = \mathcal{A}(D_k, S_k, N_k), \qquad \Delta w_k = \mathcal{U}(w_t, D_k \cup G_k), \tag{6}$$

where S_k are the local columns and types, N_k the local number of synthetic records and $\mathcal{U}$ the local training procedure. Only the model update Δw_k and approved aggregate diagnostics leave the ~~healthcare~~ facility. A coordinator aggregates the

TABLE III

PARAMETERS OF THE CONTROL LAYER, THE VALUES USED IN THE EXPERIMENTS AND THE REASON FOR EACH VALUE. THE LLM-CHOSEN SETTINGS VARIANT (SECTION V) LETS THE LLM CHOOSE THE FIRST AND FOURTH INSTEAD.

Parameter	Value	Reason
Context sample ρ	20% of training records	31 to 123 records in our datasets: enough to show the main patterns of the data, while the prompt still fits the context window of every LLM tested
Window share ω	85%	Leaves room for the answer and for errors in the estimated prompt length
Batch size b	30 records	The longest answer (30 records of the 30-feature dataset) stays within the output limit of every LLM tested; a failed answer loses at most 30 records
Temperature	0.3	Favours adherence to the output format while keeping some diversity
Repair temperature	0.1	A repair should fix the syntax, not change the content
Correction rounds c	2	Bounds the cost of a batch that keeps failing
Top-up rounds $t_{\max}$	50	$50 \times 30 = 1{,}500$ records, more than twice the largest N (614)
Empty-answer limit $u_{\max}$	5	Five consecutive empty answers indicate a provider failure rather than a sampling accident

updates as in FedAvg [44], [45],

$$w_{t+1} = w_t + \sum_{k \in \mathcal{K}_t} \frac{n_k}{\sum_{j \in \mathcal{K}_t} n_j} \Delta w_k, \qquad n_k = |D_k|, \tag{7}$$

preferably under secure aggregation [46]. The weights count *real* records, $|D_k|$ rather than $|D_k \cup G_k|$, so a ~~healthcare~~ facility gains no extra weight by generating more records (Figure 3). Unlike SiloFuse, which trains a shared generator across sites [26], this design needs no generator training across ~~healthcare~~ facilities, and each ~~healthcare~~ facility can reject its synthetic records when paired local validation shows no gain. The design is not, by itself, a privacy guarantee: with a remote LLM the prompt content leaves the ~~healthcare~~ facility, so sensitive deployments need a locally served model or a data-processing agreement, and both synthetic records and model updates can leak information [47]. Our experiments evaluate the generation step one ~~healthcare~~ facility at a time, ~~which is the setting~~ as the next section describes; a multi-facility evaluation is future work.

V. EXPERIMENTAL SETUP

This section describes the data, ~~the protocol, the configurations and the measures with which~~ protocol, configurations and measures used to answer RQ1 to RQ4~~are answered~~.

A. Datasets

~~Ten~~ We use ten public classification datasets ~~were selected~~ (Table IV): six widely used benchmarks, five from the UCI repository [48]–[52] and the Pima Indians diabetes data [53], and four further UCI datasets that are smaller, have missing values or mixed types, or are imbalanced [54]–[57]. Each

TABLE IV

EVALUATION DATASETS. n: RECORDS; d: FEATURES; k: CLASSES; $n_{\min}$: RECORDS IN THE SMALLEST CLASS; $n^\star$: MINIMUM SAMPLE SIZE BY THE FIRST CRITERION OF RILEY ET AL. [2] (SHRINKAGE 0.9, $d(k-1)$ PARAMETERS, ANTICIPATED R^2_{CS} OF 15% OF ITS MAXIMUM); RATIO: $n^\star / n_{\mathrm{tr}}$, WITH $n_{\mathrm{tr}} = 0.8n$ THE TRAINING SIZE.

Dataset	n	d	k	$n_{\min}$ (%)	$n^\star$	Ratio
Heart disease	303	13	5	13 (4.3)	10,201	42.1
Breast cancer	569	30	2	212 (37.3)	2,302	5.1
Heart failure	299	12	2	96 (32.1)	946	4.0
Diabetes	768	8	2	268 (34.9)	621	1.0
Liver disease	583	10	2	167 (28.6)	808	1.7
Parkinson's	195	22	2	48 (24.6)	1,851	11.9
Breast cancer (C.)	116	9	2	52 (44.8)	677	7.3
Hepatitis	155	19	2	32 (20.6)	1,688	13.6
Kidney disease	400	24	2	150 (37.5)	1,840	5.8
Thoracic surgery	470	16	2	70 (14.9)	1,606	4.3

For heart disease, $n^\star$ applies the binary criterion to the smallest class versus the rest and is an approximation. Parkinson's contains several recordings per subject, so its effective number of independent records is lower and its ratio higher than shown.

has fewer training records than the minimum sample size recommended for developing a prediction model for its task by the criteria of Riley et al. [2]. Together they span the factors that make small data a challenge for the classifier: 116 to 768 records, 8 to 30 features, two to five classes, a smallest class from 4.3% to 44.8% of the records and up to 10.1% missing values. We keep the original five-class heart-disease target, code the targets of the four further datasets as 0/1 and their binary categories as 0/1, and impute missing values with the mean for numerical and the mode for categorical features. The control-layer variants, the LLM variants and the sensitivity study use the six original datasets; the main configurations and all established generators use all ten.

B. Protocol

Measuring whether synthetic records help a classifier requires real records that the generator has never seen. Each dataset is therefore divided into a training set, given to the generator, and a test set of 20% of the records, used only to score classifiers. Because a test set of 24 to 154 records makes a single division unreliable, the division is repeated ten times with different random seeds, and means and standard deviations are reported. In each repetition, the number of synthetic records N equals the number of training records, so augmentation doubles the training set; because records are requested in batches of at most $b = 30$, a generator that falls short does so on small requests, not because N is large. Ten classifiers covering the main model families are trained: logistic regression (LR), support vector machine (SVM), k-nearest neighbours (kNN), Gaussian naive Bayes (NB), decision tree (DT), random forest (RF), gradient boosting (GB), AdaBoost, XGBoost and a multilayer perceptron (MLP). Using several families avoids conclusions tied to one learner, since interpolated records can help a nearest-neighbour classifier and harm a boosted ensemble. Each classifier is trained in three setups, (1) on real records only, (2) on synthetic records only and (3) on real plus synthetic records, and all

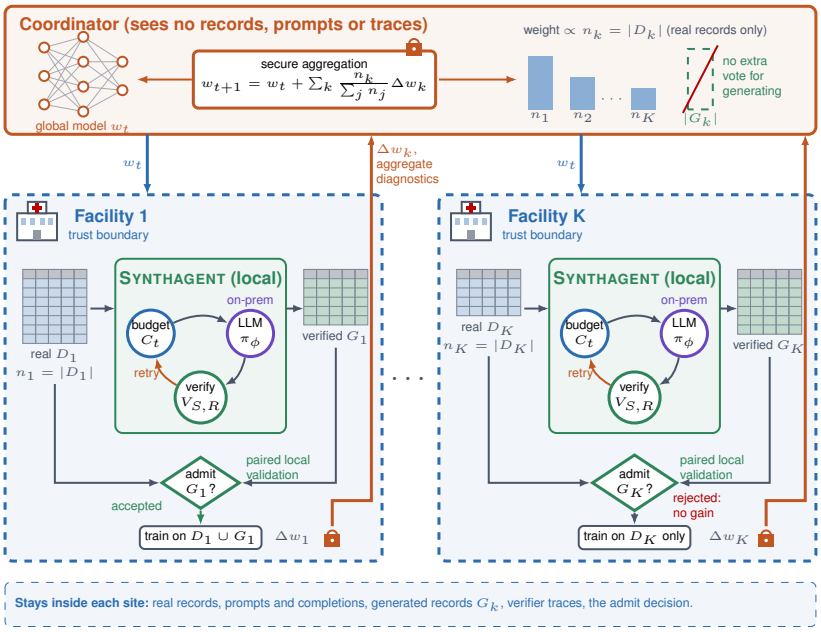


Fig. 3. Deployment across healthcare facilities. Each ~~healthcare~~ facility runs SYNTHAGENT (context selection, a locally served LLM, validation and correction) inside its own infrastructure, then decides with paired local validation whether to use the synthetic records G_k (Facility 1) or to train on real data only (Facility K). Only the model update Δw_k and approved aggregate diagnostics leave the ~~healthcare~~ facility. The coordinator weights each update by the number of real records $n_k = |D_k|$ (Equation (7)), so generating more records gives a ~~healthcare~~ facility no extra weight.

three are scored on the same test set. XGBoost failed on liver disease in every repetition because it requires classes coded from 0, whereas this dataset codes them as $\{1, 2\}$, so 99 of the 100 ~~classifier–dataset~~ classifier-dataset combinations are reported.

C. Configurations

Table V lists the ~~34~~ 36 configurations. A configuration is included in the analysis only if it completed all its datasets and ten repetitions. The *reference* configuration is SYNTHAGENT with its default settings and GPT-4o mini, the least expensive LLM; every other configuration differs from it in one respect. The *direct-prompting* baseline uses the same LLM, prompts and context, but none of the control layer~~, the~~. The *structured-output* baseline adds to it the provider's strict JSON-schema mode, which constrains decoding to the column names, numerical types and observed categories of the real data~~, and the~~. The *CLLM* baseline follows direct prompting with the curation step of CLLM [8], which keeps only the records that an XGBoost model trained on the real data learns with confidence of at least 0.75 and aleatoric uncertainty of at most 0.2 (the Data-IQ criteria [58]). Eight control-layer variants change one component or repeat the reference run, and three LLM variants replace GPT-4o mini, forming a 2×2 design that crosses model generation (4o versus 4.1) with model size (mini versus full). A second provider, Anthropic, adds three LLMs under the reference configuration, Claude Haiku 4.5, Claude Sonnet 4.6 and Claude Sonnet 5.5, and repeats both baselines with Claude Haiku 5.5, whose reference run is reported separately in Section VI-A. The Claude models run without extended reasoning and at their default temperature, which the newest of them do not allow to be changed, whereas the OpenAI models use the temperature of Table III. Token counts are compared only within a provider, because the tokenizers differ. Six sensitivity variants change one parameter of Table III at a time. ~~Eight~~ Ten established generators cover the other families, all with the default settings of their reference implementations and the same divisions: oversampling (SMOTE [4], ADASYN [59], Borderline-SMOTE [60]), statistical (a Gaussian copula [61] and a discrete Bayesian network learned by hill climbing [20]), tree-based (sequential CART synthesis [62] and adversarial random forests, ARF [63]) and deep (CTGAN~~and~~, TVAE [5] and TabDDPM [16]). In total, the configurations give 2,~~600 dataset–configuration~~ 800 dataset-configuration repetitions and up to ~~78~~84,000 classifier fits.

Three of the control-layer variants change nothing in this setting: with no user rules, disabling post-processing removes a step that does nothing, the ~~context-join option affects only a legacy fidelity score~~context join does not change the records, and disabling type rules cannot change the output because the validator never found a type violation with GPT-4o mini. Together with the reference and a rerun of it that only adds per-request logging, they are five independent runs of the same pipeline, and their spread estimates the variation between repeated runs, which we call the *noise floor*: a difference between two configurations that is smaller than this spread is not evidence of an effect.

D. Measures

Each research question ~~is answered with~~ has its own measures, all computed on every repetition: ~~reliability~~ integrity for RQ1, fidelity and privacy for RQ2, and utility for RQ3 and RQ4, with cost reported throughout. ~~*Reliability*~~*Integrity* is measured by the ~~number of~~ records delivered against ~~the number configured, the number of malformed responses, the percentage of~~ those requested, the non-empty values, the ~~corrections made and the cost in tokens~~malformed responses and the corrections. *Fidelity* compares the synthetic records

TABLE V

THE ~~34~~ 36 EVALUATED CONFIGURATIONS. ALL LLM CONFIGURATIONS USE GPT-4O MINI, EXCEPT THE LLM VARIANTS AND THE BASELINES REPEATED WITH CLAUDE HAIKU 5.5. *RUN ON ALL TEN DATASETS; THE OTHERS ON THE SIX ORIGINAL ONES.

Group	Configurations	No.	Purpose
Baselines (RQ1)	Direct prompting and structured outputs, with GPT-4o mini* and Claude Haiku 5.5; CLLM*	5	Same LLM without the control layer: free or schema-constrained decoding, or a curation filter
Reference	SYNTHAGENT, default settings*	1	Point of comparison for all others
Repeated runs (RQ1)	No type rules; no post-processing; no context join; logged rerun	4	Change nothing in this setting; estimate the variation between runs
Design choices (RQ1)	Dense context; diverse context; LLM-chosen settings; range enforcement	4	Effect of context selection, of letting the LLM set ρ and the temperature, and of clipping values to the observed ranges
Sensitivity (RQ1)	ρ of 10% and 40%; b of 15 and 60; temperature 0 and 1.0	6	Effect of each fixed parameter
LLM variants (RQ4)	GPT-4.1 mini; GPT-4o; GPT-4.1*; Claude Haiku 4.5; Claude Sonnet 4.6*; Claude Sonnet 5.5	6	Effect of model generation, size and provider
Established generators* (RQ4)	SMOTE; ADASYN; Borderline-SMOTE; Gaussian copula; Bayesian network; CART; ARF; CTGAN; TVAE; TabDDPM	10	Generators fitted to the real records
Total		36	

alone with the real training records over the numerical features $\mathcal{J}$:

$$F_{\mathrm{JSD}} = 100\Big(1 - \tfrac{1}{|\mathcal{J}|}\textstyle\sum_{j\in\mathcal{J}} D_{\mathrm{JS}}(D_j, G_j)\Big),$$
$$F_{\mathrm{corr}} = 100\big(1 - \mathrm{mean}\,|\Sigma_D - \Sigma_G|\big), \tag{8}$$

where D_{JS} is the ~~Jensen–Shannon~~ Jensen-Shannon divergence between real and synthetic distributions of a feature and Σ_D, Σ_G are the correlation matrices; both scores equal 100 for identical distributions. A classifier two-sample test (C2ST) [38] trains a random forest to separate real from synthetic records; an accuracy close to 0.5 means ~~that~~ they are hard to tell apart. ~~For the established generators and direct prompting we also report the percentage of *valid* records, whose values lie in the observed ranges and categories and are integers where the real feature is, a stricter criterion than the column-and-type checks of the control layer.~~ *Privacy* is measured by a membership-inference attack that scores each real record by its distance to the nearest synthetic record and reports the ROC AUC between training and test records, where 0.5 is chance, and by the share of *near-copies*, synthetic records within 0.05 of a real record after min-max scaling. *Utility* is the paired change, for metric $m \in \{\mathrm{accuracy}, \mathrm{F1}\}$, dataset d, classifier c and repetition h,

$$\Delta m_{d,c,h} = m\big(D_{d,h} \cup G_{d,h};\, c\big) - m\big(D_{d,h};\, c\big), \tag{9}$$

TABLE VI

RESULTS AT A GLANCE OVER THE TEN DATASETS: RANGES OVER THE THREE LLMS OF SYNTHAGENT, THE THREE OTHER LLM-BASED GENERATORS AND THE TEN ESTABLISHED GENERATORS. ONLY SYNTHAGENTHAS NO INTEGRITY VIOLATIONS, AND IT REACHES THE LARGEST MEDIAN F1 GAIN.

	SYNTHAGENT	Other LLM-based	Established
Integrity violations	none	13-22% of records missing	no LLM
Malformed responses	repaired	batches lost	-
Near-copies of real records (%)	6-9	4-7	1-19
Membership-attack AUC	0.51-0.52	0.51-0.52	0.50-0.57
Fidelity (JSD)	72-77	66-72	62-86
Median F1 change (pp)	-0.69 to $+1.43$	-0.68 to $+0.24$	-2.15 to $+0.11$
Tokens per delivered record	-17% vs. direct	-	no LLM

with both models scored on the same test set. We use binary F1 for the binary tasks and class-weighted F1 for heart disease, and report changes in percentage points (pp).

E. Statistical Analysis

Repetitions and classifiers are not independent datasets, so they are averaged within each dataset and the ~~six~~ datasets are treated as matched blocks. Configurations are compared with the Friedman test,

$$\chi^2_F = \frac{12M}{Q(Q+1)}\sum_{a=1}^{Q}\bar{\mathrm{rk}}_a^{\,2} - 3M(Q+1), \tag{10}$$

where $\bar{\mathrm{rk}}_a$ is the average rank of configuration a over ~~$M = 6$ datasets~~ M datasets (six or ten) and Q the number of configurations, followed by two-sided Wilcoxon signed-rank tests of the best-ranked configuration against every other, with Holm correction [64], [65]. With ~~six~~ few blocks the tests have limited power, so effect sizes and the noise floor are reported alongside p-values. As a sensitivity analysis, the comparison of LLMs is repeated with the 60 ~~dataset–division~~ matched dataset-division pairs as blocks~~: all configurations use the same divisions, so the pairs are matched, but divisions of one dataset~~; these share records, so ~~these blocks are not independent and~~ the dataset-level tests remain the primary analysis.

VI. RESULTS

Table VI summarizes the results; the subsections detail them by research question.

A. RQ1: Control Layer and ~~Reliability~~Integrity

Finding 1. The control layer ~~and the reliability of the generated datasets are directly related~~ensures integrity: without it, every run of the same LLM ~~never delivered the configured number of~~ missed or added records; with it, ~~every dataset was complete~~no logged run had an integrity violation, at 1.9% more tokens and with unchanged ~~quality. Table ?? compares direct prompting with the reference on the same prompts, context and divisions. Direct prompting delivered on average~~ fidelity. Figure 4 shows the records delivered per dataset. On the six original datasets, direct prompting delivered 87.6% of the configured records, fell short in 50 of 60 runs and overshot in the other 10, and lost an entire batch to each of ten

malformed responses. ~~delivers exactly N records because its completion loop requests the missing ones and truncates any excess, and its parser recovers malformed responses instead of discarding them.~~ Fidelity, privacy and utility were the same for both configurations (JSD 72.7; attack AUC 0.517 versus 0.526; F1 change -0.04 versus -0.14 pp, within the noise floor below), so the difference lies ~~not in the records but in whether the datasetthat was configured is delivered~~in the integrity of the dataset.

Structured outputs, the ~~option offered by LLM providers~~providers' option to force every response into a fixed format, do not replace the control layer (~~Table ??~~Figure 4). Every response had the declared columns and types, but the LLM returned on average 77.7% of the configured records and never the full dataset, at 21% more tokens than SYNTHAGENT. Forcing the format also weakened the relationships between features (correlation fidelity 63.0 versus 72.6) and made the records easier to tell from real ones (C2ST 0.943 versus 0.878), while utility did not differ (Friedman $p = .311$ for F1). A fixed format therefore guarantees the shape of each record, but not the ~~size of the dataset~~number of records, which is what the completion loop provides. The same holds for the second provider: with Claude Haiku 5.5, direct prompting delivered ~~on average over datasets~~ 67.1% of the configured records, with 294 malformed responses, and structured outputs 82.9%~~, from 37.6% to 107.6% depending on the dataset~~. Under the control layer, the same model declined in plain text to imitate the patient records in 39 of the 60 repetitions of a first pass; SYNTHAGENT logged each refusal instead of returning ~~an incomplete~~ a smaller dataset, but ~~completed~~ finished only heart failure, so this run is not included in the comparisons.

These comparisons isolate ~~what~~ the control layer~~contributes~~'s contribution, because the LLM, ~~the prompts and the~~ prompts and context are the same in each pair. Its contribution is ~~reliability~~integrity, at no cost in utility: with GPT-4o mini~~on the ten datasets~~, SYNTHAGENT ~~delivered every configured record against 86% for direct prompting and 78% for structured outputs~~ had no integrity violation in any logged run, while direct prompting missed 13% of the records and structured outputs 22% over ten datasets, and turned malformed responses into records instead of losing them, while the F1 change of the two did not differ (median difference $+0.08$ pp, Wilcoxon $p = .92$). ~~The gains in utility~~ On the 40 paired runs with logged counts, the difference in integrity is significant (exact McNemar test, $p < 10^{-11}$), and SYNTHAGENTneeded 17% fewer tokens per delivered record than direct prompting and 40% fewer than structured outputs (Wilcoxon $p < 10^{-6}$), because the baselines pay for records they lose. The results therefore come from the ~~LLM that~~ two together: integrity is due to the control layer~~makes usable, which is why they grow with stronger LLMs (Section VI-D), and not from the checks themselves, none of whose variants changed utility (below~~, which achieved it with every LLM evaluated while no LLM achieved it alone, and the utility gains come from the LLM's knowledge, which the control layer turns into a usable dataset (Section VI-D).

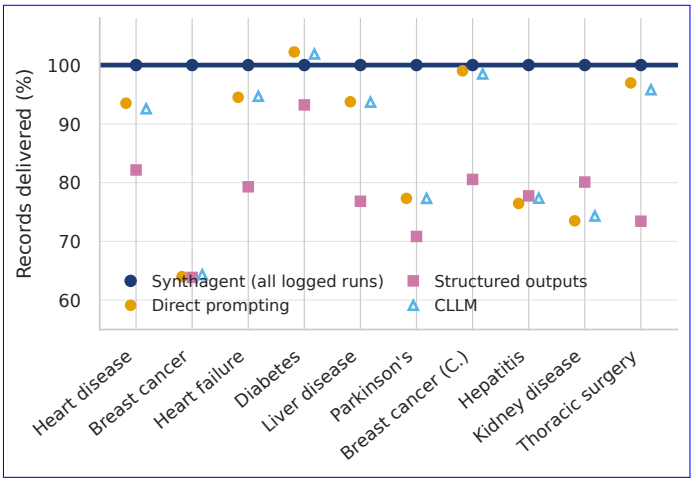


Fig. 4. ~~Direct prompting, structured outputs and~~ Only SYNTHAGENT has no integrity violations (RQ1)~~, all with GPT-4o mini, over ten divisions. Deliv.:~~ ; the LLM baselines miss or add records ~~returned~~ on every dataset. Records delivered as a percentage of ~~the configured number N (range~~ those requested, averaged over ten divisions~~for direct prompting); Malf. : malformed responses; ΔF1: mean F1 change (pp)~~Baselines use GPT-4o mini; ~~bold: best per dataset.~~ SYNTHAGENT ~~delivered 100% in~~ points are every run with a logged count (GPT-4o mini and GPT-4.1 on the four further datasets, Claude Sonnet 4.6 on all ten).

The design choices inside the control layer matter much less than the control layer itself (~~Table ??~~Figure 5). The five repeated runs of the same pipeline differ by up to 0.5 JSD points, 2 correlation points and 0.7 pp of utility, which is the noise floor. Dense context stays inside it; diverse context slightly improves fidelity ~~(JSD 73.27, correlation 73.72) at the cost of~~ at a slightly higher attack AUC (0.544); ~~letting the LLM choose its own settings changes nothing but costs~~ LLM-chosen settings cost 57% more tokens; and ~~clipping values to the observed ranges lowers JSD~~ range enforcement lowers JSD fidelity by 0.9 points~~, just beyond the noise floor, without improving utility~~. The Friedman tests find no difference in utility between the variants ($p = .843$ for accuracy, $p = .872$ for F1). The parameters of Table III behave the same way: changing the context sample, the batch size or the temperature leaves utility within the noise floor ($p = .986$ ~~and~~ accuracy, $p = .178$ F1), but it changes cost and fidelity. Halving the context sample saves 27% of the tokens and doubling the batch size 39%, ~~at slightly lower correlation fidelity,~~ whereas smaller batches cost 73% more. A temperature of 1.0 ~~is the one change with a large effect: it raises JSD~~ raises JSD fidelity by 3.2 points and correlation fidelity by 8.7 points ~~and makes the records harder to detect~~ (C2ST 0.799 against 0.878) ~~,~~ at no cost in utility, privacy or tokens, ~~which suggests that~~ so the default of 0.3 is ~~too~~ conservative for fidelity.

The corrections ~~are what allow~~ make a less disciplined LLM ~~to be used~~usable. GPT-4o mini produced no type violation, and the local parser fixed every defect in its responses. GPT-4.1 mini, the fastest LLM, produced on average 7.4 type violations per repetition on breast cancer and one malformed

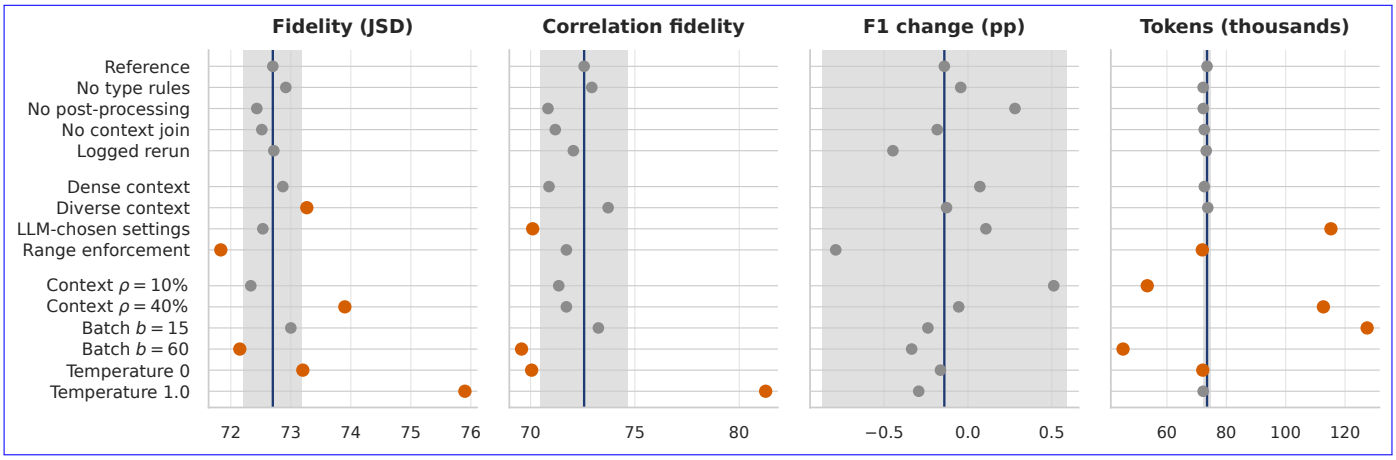


Fig. 5. ~~Control-layer variants~~ No design choice or parameter moves utility beyond the noise floor; the parameters change only cost and ~~sensitivity study~~ fidelity (RQ1)~~, all with~~ , GPT-4o mini~~on the~~ , six original datasets. The first five rows ~~run~~ rerun the same pipeline~~and define the~~ ; grey band: noise floor ~~. Seconds and tokens are means per repetition. Bold: best value among the LLM configurations; lower is better for C2ST~~(reference, ~~for the distance of the attack AUC from 0.5~~blue line, ~~and for cost~~± their spread); orange: outside it.

TABLE VII
THE SYNTHETIC RECORDS FOLLOW THE REAL DISTRIBUTIONS AND KEEP THE ATTACK NEAR CHANCE ON EVERY DATASET (RQ2, RQ3).
REFERENCE CONFIGURATION (GPT-4O MINI) PER ORIGINAL DATASET, ~~AVERAGED~~ OVER TEN REPETITIONS AND ALL CLASSIFIERS~~(RQ2, RQ3).~~ ; JSD AND CORRELATION ~~ARE~~ ON A ~~0–100~~ 0-100 SCALE.

	Fidelity and privacy				Acc. (%)		Change (pp)	
Dataset	JSD	Corr.	C2ST	AUC	Real	Aug.	Acc.	F1
Breast	77.2	74.5	0.911	0.550	95.1	94.4	−0.66	−0.62
Diabetes	80.2	44.0	0.897	0.504	74.2	74.1	−0.03	+2.39
Heart dis.	65.2	76.6	0.893	0.507	56.6	53.4	−3.15	−0.97
Heart fail.	71.2	81.2	0.892	0.534	74.1	71.6	−2.53	−1.00
Liver	71.8	73.8	0.883	0.495	67.4	67.6	+0.19	+0.13
Parkinson's	70.6	85.3	0.789	0.567	84.9	84.0	−0.97	−0.77
Mean	72.7	72.6	0.878	0.526	75.4	74.2	−1.19	−0.14

response; these were sent back for correction instead of entering the dataset, and its datasets ~~were complete~~ had no integrity violations and gave the best accuracy change of ~~all configurations (Table VIII~~the OpenAI LLMs (Section VI-D).

B. RQ2: Fidelity and Privacy

Finding 2. The generated records follow the real distributions reasonably well without revealing the real records, but they remain distinguishable from them (Table VII). Fidelity averages 72.7 out of 100, from 65.2 on ~~the five-class heart-disease dataset~~heart disease, the furthest below its required sample size, to 80.2 on diabetes, the closest~~to it~~. The membership-inference attack stays near chance (AUC 0.526 on average, at most 0.567), which is a diagnostic for one attack rather than a formal guarantee. The ~~C2ST accuracy of 0.878 shows that a~~ records are also not interpolations of training patients: 7.8% are near-copies, against 17-19% for the oversampling methods. A classifier can still separate real from synthetic records ~~, so the~~ (C2ST 0.878), so synthetic records complement real ones rather than replace them.

C. RQ3: Utility for Classifiers Trained on Small Data

Finding 3. The generated records help the classifiers that fail on the minority class and slightly harm those that are already strong, so their effect depends on the dataset and the classifier. ~~Classifiers trained on real data alone already vary widely: on the 195-record Parkinson's dataset, MLP accuracy is 72.3 ± 18.2% while tree ensembles exceed 92%, and on heart failure~~On heart failure, for example, SVM and kNN reach over 60% accuracy with an F1 of almost zero because they rarely predict the minority class. With the reference configuration, F1 improves in 26 of the 59 ~~classifier–dataset~~ classifier-dataset combinations and decreases in 31 (Figure 6); the largest gains occur where real-only training fails, such as kNN on heart failure (+15.3 F1 points)~~and MLP on Parkinson's (+8.2 accuracy points). No single~~ . No combination is significant after Holm correction~~, and the change varies widely between divisions~~. On average, F1 is unchanged with GPT-4o mini (−0.14 pp) and ~~increases by 1.09 pp with GPT-4.1, which is consistent with the bound set by information theory: synthetic records cannot carry more information about the task than the real records and the generator's knowledge together [6, Sec. 2.8]~~gains more with stronger LLMs (Section VI-D).

D. RQ4: Comparison with Established Generators and LLMs

~~With a strong LLM,~~ **Finding 4.** SYNTHAGENT ~~gives~~ achieves the largest median F1 gain among the generators compared, and the LLM is the ~~factor of the control layer's configuration~~ setting that most changes the results. Among the LLMs (Table VIII), the Claude models give the strongest results in our runs: Claude Sonnet 4.6 reaches the highest fidelity (79.07) and the largest gains (+0.40 pp of accuracy, +2.39 pp of F1), and all three Claude models exceed the best OpenAI model in F1. This comparison ~~between providers~~ is suggestive rather than conclusive~~:~~ , since the Claude models ran at their default temperature and the OpenAI models at

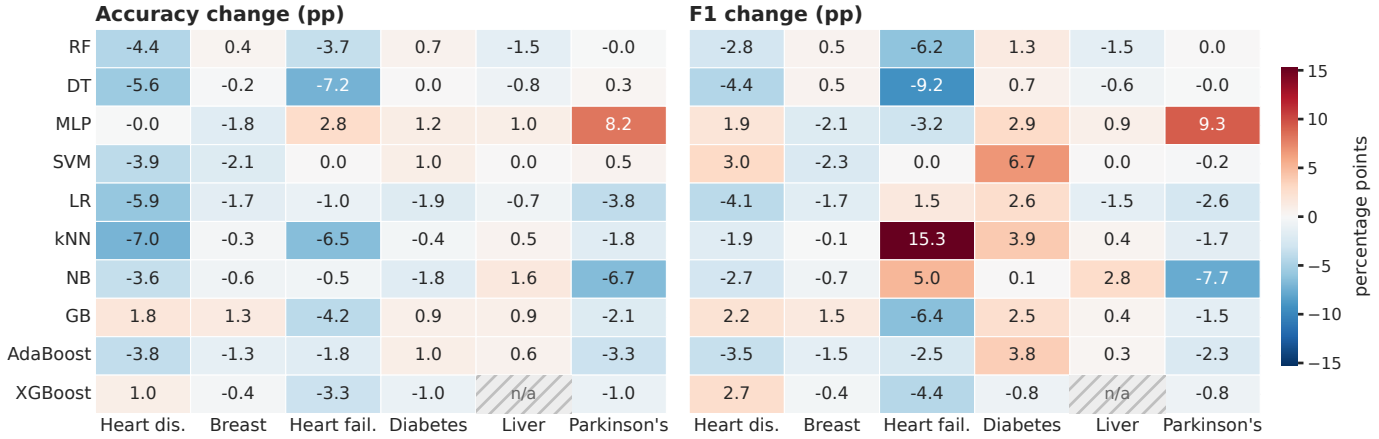


Fig. 6. The synthetic records help the classifiers that fail on real data alone and slightly harm strong ones (RQ3). Change from real-only to real-plus-synthetic training~~for the~~ , reference configuration~~(RQ3)~~, per classifier and dataset. ~~Hatched cells (n/a) were~~ ; hatched: not evaluated. ~~The same synthetic data can help one classifier and harm another.~~

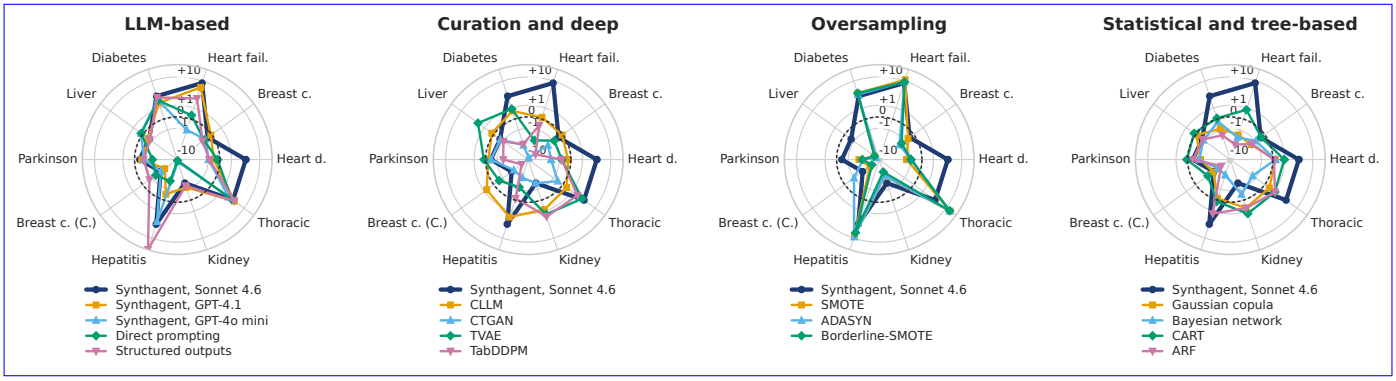


Fig. 7. SYNTHAGENTgains most where real-only training is weak; oversampling swings most (RQ4). F1 change (pp) per dataset for every generator, one panel per group, each against SYNTHAGENTwith Claude Sonnet 4.6 (dark blue); symmetric logarithmic radius (linear within ±1 pp); dashed circle: no change.

TABLE VIII
UNDER THE CONTROL LAYER, NO LLM PRODUCED AN INTEGRITY VIOLATION IN ANY LOGGED RUN; THE CLAUDE MODELS GIVE THE HIGHEST FIDELITY AND F1 GAIN (RQ4). REFERENCE CONFIGURATION; SECONDS AND TOKENS ARE MEANS PER REPETITION, TOKENS COMPARABLE ONLY WITHIN A PROVIDER.

LLM	JSD	Corr.	ΔAcc.	ΔF1	Sec.	Tokens
GPT-4o mini	72.70	**72.57**	−1.19	−0.14	378.7	73,510
GPT-4.1 mini	73.60	62.82	−0.23	+0.92	**168.5**	**72,069**
GPT-4o	73.95	64.32	−0.50	+0.25	255.2	80,046
GPT-4.1	75.72	63.30	−0.44	+1.09	201.7	80,652
Claude Haiku 4.5	74.85	69.88	−0.34	+1.55	253.5	78,020
Claude Sonnet 4.6	**79.07**	70.77	+0.40	+2.39	373.5	**75,504**
Claude Sonnet 5.5	79.00	63.78	+0.10	+1.62	205.5	86,074

0.3~~, and a temperature of 1.0 alone raised the JSD of GPT-4o mini by 3.2 points~~. Within OpenAI, GPT-4.1 gives the highest fidelity (75.72) and F1 gain (+1.09 pp)~~and GPT-4.1 mini the best accuracy change at the lowest generation time~~, while the three stronger OpenAI models preserve correlations ~~between features~~ less well (62.8 to 64.3, against 72.6 for GPT-4o mini). The control layer ~~completed~~ delivered every dataset with every ~~one of these LLMs~~LLM, and the differences between them are not significant across six datasets ($p = .572$ and $p = .615$ within OpenAI, $p = .377$ over all seven, for accuracy and F1). With the 60 divisions as blocks, the seven LLMs differ ($p \leq .002$), and Claude Sonnet 4.6 gives a higher F1 than each of the others after Holm correction ($p \leq .039$).

~~LLMs under the reference configuration (RQ4). Seconds and tokens are means per repetition; tokens are comparable only within a provider. All datasets were complete.~~

Over the ten datasets (Table IX, Figure 7), no generator is best on every measure~~, and the generators fall into three groups~~. The tree-based and statistical generators reproduce the real distributions most closely (JSD 85.0 to 86.0), and CART's records are as hard to tell from real ones as real records are (C2ST 0.465), but ~~CART also has the highest membership-attack~~ have the highest attack AUC (0.565)~~, because its records stay close to the real ones. The oversampling methods lower accuracy on most datasets~~ . High fidelity does not imply utility: the Bayesian network matches the copula in fidelity (JSD 85.1) but improves F1 on only one dataset. Oversampling lowers accuracy (median −2.3 to −3.1 pp) and ~~vary~~ varies the most in F1 (~~standard deviation~~

TABLE IX

NO GENERATOR IS BEST ON EVERY MEASURE; SYNTHAGENTGIVES THE LARGEST MEDIAN F1 GAIN (RQ4). THE ~~FOURTEEN~~ SIXTEEN GENERATORS OVER THE TEN DATASETS~~(RQ4)~~. JSD, C2ST AND ATTACK AUC: MEANS OVER DATASETS. ΔACC. AND ΔF1: MEDIAN $\pm$ STANDARD DEVIATION OVER DATASETS OF THE CHANGE (PP) FROM REAL-ONLY TRAINING. IMPROVED: DATASETS WITH A POSITIVE F1 CHANGE (%). BOLD: BEST VALUE; FOR C2ST AND ATTACK AUC, CLOSEST TO 0.5.

	Fidelity and privacy			Utility		
Generator	JSD	C2ST	Attack AUC	ΔAcc. (pp)	ΔF1 (pp)	Improved (%)
Synthagent, GPT-4o mini	71.8	0.878	0.524	-0.99 ± 2.00	-0.69 ± 3.80	40
Synthagent, GPT-4.1	73.7	0.874	0.521	-0.68 ± 1.08	-0.45 ± 2.98	30
Synthagent, Sonnet 4.6	77.4	0.838	0.514	-0.37 ± 1.04	$+1.43 \pm 3.51$	50
Direct prompting	71.7	0.883	0.514	-1.47 ± 2.31	-0.47 ± 3.95	40
Structured outputs	70.8	0.934	0.506	-0.97 ± 0.98	-0.68 ± 7.01	40
CLLM	66.4	0.893	0.516	$+0.07 \pm 0.37$	$+0.24 \pm 0.64$	70
SMOTE	65.8	0.830	0.524	-2.25 ± 7.46	-1.01 ± 7.43	40
ADASYN	61.9	0.851	0.518	-3.08 ± 7.00	-0.95 ± 7.23	40
Borderline-SMOTE	65.2	0.862	0.519	-2.76 ± 6.87	-1.20 ± 7.25	40
Gaussian copula	85.2	0.541	0.522	-0.26 ± 1.31	-0.49 ± 1.10	30
Bayesian network	85.1	0.656	0.516	-0.75 ± 1.84	-0.97 ± 1.51	10
CART	**86.0**	**0.465**	0.565	-0.17 ± 0.66	$+0.11 \pm 0.82$	**80**
ARF	85.0	0.697	0.532	-0.37 ± 1.41	-0.53 ± 2.27	40
CTGAN	79.2	0.917	**0.500**	-1.41 ± 2.05	-2.15 ± 3.01	0
TVAE	67.7	0.866	0.515	-0.22 ± 1.27	-0.20 ± 1.80	50
TabDDPM	63.6	0.883	0.525	-0.60 ± 1.94	-0.90 ± 3.13	20

SD above 7 pp). ~~Among the LLM-based generators,~~ SYNTHAGENT with Claude Sonnet 4.6 gives the largest median F1 gain of all ~~fourteen~~ sixteen generators (+1.43 pp) and improves F1 on half of the datasets, more often than any other LLM configuration except CLLM. The CLLM filter changes the data least: it gives the best median accuracy change (+0.07 pp) and the smallest variation in F1, and improves F1 on 70% of the datasets, and CART on 80%, ~~but~~ both with small gains (+0.24 and +0.11 pp)~~. Unlike , which delivered every configured record, CLLM delivered 86% of them, because it discards records, as did direct prompting(86%) and structured outputs (78%)~~; but CLLM missed 13% of the requested records, the shortfall of direct prompting, whose output it filters. The generators differ across the ten datasets (Friedman $p < .001$ for accuracy, ~~$p = .043$~~ $p = .004$ for F1), with SYNTHAGENT and Claude Sonnet 4.6 ranked first in F1 and CLLM in accuracy; after Holm correction, CLLM has a higher accuracy than ~~the three oversampling methods~~ SMOTE and Borderline-SMOTE, and SYNTHAGENT with Sonnet 4.6 a higher accuracy than Borderline-SMOTE, and ~~no pairwise~~ a higher F1 ~~difference is significant~~than TabDDPM on every dataset ($p = .029$).

~~F1 change (pp) per dataset for five representative generators (RQ4); the dashed circle is no change. with Claude Sonnet 4.6 gains most where real-only training is weak (heart failure, heart disease, thoracic surgery), and SMOTE has the largest gains and losses.~~

VII. DISCUSSION AND LIMITATIONS

The ~~objective of this work was to determine whether makes LLM-based SDG a reliable and useful option for small healthcare datasets. Reliability is achieved, and it is achieved by~~ results come from the LLM and the control layer ~~: the same LLM , prompted directly, never delivered the configured dataset , whereas completed it, recorded its cost and caught the failures of a less disciplined model~~ together: the LLM brings domain knowledge, and the control layer turns it into a dataset that satisfies its integrity constraints, which no LLM did alone, at almost no extra cost (RQ1). ~~Usefulness~~ The usefulness is real but bounded ~~and depends on the LLM: the synthetic records preserve the distribution of the real data without revealing real records (RQ2), they help the classifiers that fail on the minority class (RQ3), and with a strong LLM they give the largest median gain among the generators compared, while a weak LLM can lose as much as direct prompting does (RQ4). The alternatives trade one property for another: curation is more stable but discards records, and tree-based generators fit the real distributions more closely by staying close to the real records.~~ (RQ2-RQ4). SYNTHAGENT is not the best choice in three situations. With a weak LLM it does not improve on direct prompting in utility, and on kidney disease, with mixed types and the most missing values, GPT-4o mini lost 10.3 pp of F1 with or without the control layer, so a weak LLM should not be used for such data. When the goal is ~~to reproduce~~ the closest fit to the real distribution~~as closely as possible~~, tree-based and statistical generators are better, ~~but they do so by staying close to the real records, which raises the~~ at a higher membership-attack risk. And when stability matters more than gain, the CLLM filter changes the data least, at the price of ~~an incomplete dataset~~fewer records than configured; since the two address different failures, combining the control layer with such a filter is a natural extension. No generator ~~, however,~~ improves every dataset~~, which is consistent with the data-processing inequality [6, Sec. 2.8]~~ : SDG cannot replace missing patients~~, and its benefit is bounded by the knowledge the generator brings~~ [6, Sec. 2.8] . In practice, the ~~choice of~~ LLM matters most, ~~temperature is the one parameter that changes fidelity, cost is better measured in tokens than in time (generation time varied by a factor of~~

1.7 between repeated runs), and the effect of and synthetic records should be checked per classifierbefore they are used.

The conclusions are subject to study has several limitations. Ten datasets act as independent blocks, and (six for the control-layer and LLM variants, which limits variants) limit the power of the tests; , and the divisions are not stratified. The datasets are publicand widely used, so they may be part of the LLMs' training data and , so part of the fidelity may reflect memorized records rather than generation; the membership-inference records memorized by the LLMs, which the attack does not testfor this, and an evaluation on private data is ; private data are left for future work. The Parkinson's dataset contains several recordings per subject, so recordings of one person can appear subject can fall in both sets. The columns Columns and types are inferred from the full datasetrather than from the training set alone. The parameters in Table III were fixed and not tuned; the sensitivity study varies three of them one at a time, not jointly. The temperature differs between providers, and part of the Claude models' higher fidelity may reflect it (Section VI-A). does not filter synthetic records that are close to real ones: privacy VI-D). Privacy is measured with one distance-based attack, not enforced, and no result establishes differential privacy [42]. The runs Delivered counts were not logged for the GPT runs of SYNTHAGENT on the six original datasetspredate the counter of delivered records, so their size follows from the completion loop rather than from a logged count. The CLLM baseline reimplements its curation step on our direct-prompting output rather than running the original code. The corrections were exercised only by GPT-4.1 mini, and records obtained in the last correction round or in the completion loop are not validated again. Context selection is not class-stratified, so a rare class can be missing from a prompt. The LLMs come from two commercial providers and locally served models Local LLMs were not evaluated; one of them, , and Claude Haiku 5.5, declined to imitate patient records under the control layer's prompts, which shows 's refusals show that a provider's safety behaviour can block generationon healthcare data. The established generators use default settings, and TabDDPM and GReaT, which need needs a GPU, were was not included.

Synthetic healthcare records must not inform diagnosis or policy without domain review, and near-chance performance against one attackdoes not justify releasing sensitive datanor be released on the strength of one attack. The implementation, the experiment runner and the script that regenerates every table and figure are publicly available.[1]

VIII. CONCLUSION

Small healthcare datasets pose a dilemma for synthetic data generation: generators fitted to the available records Generators fitted to small healthcare datasets can only recombine the patternsthose records containtheir patterns, whereas an LLM can add adds domain knowledge but returns text rather than a dataset. SYNTHAGENT resolves the dilemma this with a control layer that turns the responses of an LLM into complete synthetic datasets with the columns and types of the real data and LLM responses into synthetic datasets without integrity violations, with a recorded cost, without training on the target data. On ten public healthcare datasets, direct prompting never delivered the configured dataset, with LLMs of two providers and even with responses forced into a fixed formatviolated integrity in every run, while SYNTHAGENT did at 1.9% more cost, and with the strongest LLM it gave never did, with fewer tokens per delivered record, and it achieved the largest median F1 gain of fourteen generators. Because repeated runs of the same pipeline vary as much as most design choices, we recommend that SDG for small data be evaluated on repeated runs. sixteen generators. Future work will evaluate locally served local LLMs, class-aware contextselection, privacy filters and the deployment across healthcare facilities outlined in of Section IV.

[1] TODO(authors): add the public repository URL required by the ICDE artifact policy.https://github.com/smartlord7/Synthagent

ACKNOWLEDGMENT

This work is funded by national funds through FCT — Foundation for Science and Technology, I.P., within the scope of the research unit UID/00326 — Centre for Informatics and Systems of the University of Coimbra, https://doi.org/10.54499/UID/00326/2025.

AI-GENERATED CONTENT ACKNOWLEDGEMENT

The authors used Claude (Anthropic) to assist with code development and with the writing of the manuscript, and take full responsibility for all content.

REFERENCES

[1] N. Rieke, J. Hancox, W. Li, F. Milletarì, H. R. Roth, S. Albarqouni, S. Bakas, M. N. Galtier, B. A. Landman, K. Maier-Hein, S. Ourselin, M. Sheller, R. M. Summers, A. Trask, D. Xu, M. Baust, and M. J. Cardoso, "The future of digital health with federated learning," *npj Digital Medicine*, vol. 3, no. 1, p. 119, 2020.

[2] R. D. Riley, K. I. E. Snell, J. Ensor, D. L. Burke, F. E. Harrell, Jr., K. G. M. Moons, and G. S. Collins, "Minimum sample size for developing a multivariable prediction model: Part II – binary and time-to-event outcomes," *Statistics in Medicine*, vol. 38, no. 7, pp. 1276–1296, 2019.

[3] Z. Qian, R. Davis, and M. van der Schaar, "Synthcity: A benchmark framework for diverse use cases of tabular synthetic data," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.

[4] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.

[5] L. Xu, M. Skoularidou, A. Cuesta-Infante, and K. Veeramachaneni, "Modeling tabular data using conditional GAN," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[6] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Hoboken, NJ: Wiley-Interscience, 2006.

[7] V. Borisov, K. Seßler, T. Leemann, M. Pawelczyk, and G. Kasneci, "Language models are realistic tabular data generators," in *International Conference on Learning Representations*, 2023. [Online]. Available: https://openreview.net/forum?id=cEygmQNOeI

[8] N. Seedat, N. Huynh, B. van Breugel, and M. van der Schaar, "Curated LLM: Synergy of LLMs and data curation for tabular augmentation in low-data regimes," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235, 2024.

[9] C. Ge, S. Mohapatra, X. He, and I. F. Ilyas, "Kamino: Constraint-aware differentially private data synthesis," *Proceedings of the VLDB Endowment*, vol. 14, no. 10, pp. 1886–1899, 2021.

[10] J. Rorseth, P. Godfrey, L. Golab, D. Srivastava, and J. Szlichta, "Recovering structure in unstructured LLM outputs," in *Proceedings of the IEEE 42nd International Conference on Data Engineering (ICDE)*, 2026, pp. 4329–4330.

[11] S. Schelter, D. Lange, P. Schmidt, M. Celikel, F. Biessmann, and A. Grafberger, "Automating large-scale data quality verification," *Proceedings of the VLDB Endowment*, vol. 11, no. 12, pp. 1781–1794, 2018.

[12] E. Breck, N. Polyzotis, S. Roy, S. E. Whang, and M. Zinkevich, "Data validation for machine learning," in *Proceedings of the 2nd SysML Conference*, 2019.

[13] R. Y. Wang and D. M. Strong, "Beyond accuracy: What data quality means to data consumers," *Journal of Management Information Systems*, vol. 12, no. 4, pp. 5–33, 1996.

[14] C. Batini, C. Cappiello, C. Francalanci, and A. Maurino, "Methodologies for data quality assessment and improvement," *ACM Computing Surveys*, vol. 41, no. 3, pp. 16:1–16:52, 2009.

[15] A. D. Lautrup, T. Hyrup, A. Zimek, and P. Schneider-Kamp, "SynthEval: A framework for detailed utility and privacy evaluation of tabular synthetic data," *Data Mining and Knowledge Discovery*, vol. 39, no. 1, p. 6, 2025.

[16] A. Kotelnikov, D. Baranchuk, I. Rubachev, and A. Babenko, "TabDDPM: Modelling tabular data with diffusion models," in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202, 2023, pp. 17 564–17 579.

[17] H. Zhang, J. Zhang, Z. Shen, B. Srinivasan, X. Qin, C. Faloutsos, H. Rangwala, and G. Karypis, "Mixed-type tabular data synthesis with score-based diffusion in latent space," in *International Conference on Learning Representations*, 2024. [Online]. Available: https://openreview.net/forum?id=4Ay23yeuz0

[18] A. V. Solatorio and O. Dupriez, "REaLTabFormer: Generating realistic relational and tabular data using transformers," *arXiv preprint arXiv:2302.02041*, 2023.

[19] J. Kim, T. Kim, and J. Choo, "EPIC: Effective prompting for imbalanced-class data synthesis in tabular data classification via large language models," in *Advances in Neural Information Processing Systems*, vol. 37, 2024.

[20] J. Zhang, G. Cormode, C. M. Procopiuc, D. Srivastava, and X. Xiao, "PrivBayes: Private data release via Bayesian networks," in *Proceedings of the ACM SIGMOD International Conference on Management of Data*, 2014.

[21] B. T. Willard and R. Louf, "Efficient guided generation for large language models," *arXiv preprint arXiv:2307.09702*, 2023.

[22] S. Geng, M. Josifoski, M. Peyrard, and R. West, "Grammar-constrained decoding for structured NLP tasks without finetuning," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023, pp. 10 932–10 952.

[23] K. Cai, X. Lei, J. Wei, and X. Xiao, "Data synthesis via differentially private Markov random fields," *Proceedings of the VLDB Endowment*, vol. 14, no. 11, pp. 2190–2202, 2021.

[24] C. Lee, J. Kim, and N. Park, "CoDi: Co-evolving contrastive diffusion models for mixed-type tabular synthesis," in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202, 2023, pp. 18 940–18 956.

[25] J. Fan, T. Liu, G. Li, J. Chen, Y. Shen, and X. Du, "Relational data synthesis using generative adversarial networks: A design space exploration," *Proceedings of the VLDB Endowment*, vol. 13, no. 11, pp. 1962–1975, 2020.

[26] A. Shankar, H. Brouwer, R. Hai, and L. Y. Chen, "SiloFuse: Cross-silo synthetic data generation with latent tabular diffusion models," in *Proceedings of the IEEE 40th International Conference on Data Engineering (ICDE)*, 2024, pp. 110–123.

[27] L. Chen, C. Wu, J. Li, S. Liu, G. Ye, and H. Chai, "TabLoft: Tabular data generation based on LLM with ordered features," in *Proceedings of the IEEE 42nd International Conference on Data Engineering (ICDE)*, 2026, pp. 1044–1056.

[28] J. Hyeong, J. Kim, N. Park, and S. Jajodia, "An empirical study on the membership inference attack against tabular data synthesis models," in *Proceedings of the 31st ACM International Conference on Information and Knowledge Management (CIKM)*, 2022, pp. 4064–4068.

[29] X. Chu, I. F. Ilyas, and P. Papotti, "Holistic data cleaning: Putting violations into context," in *Proceedings of the IEEE 29th International Conference on Data Engineering (ICDE)*, 2013, pp. 458–469.

[30] J. Qin, S. Huang, Y. Wang, J. Zhu, Y. Zhang, Y. Miao, R. Mao, M. Onizuka, and C. Xiao, "BClean: A Bayesian data cleaning system," in *Proceedings of the IEEE 40th International Conference on Data Engineering (ICDE)*, 2024, pp. 3407–3420.

[31] A. Narayan, I. Chami, L. Orr, S. Arora, and C. Ré, "Can foundation models wrangle your data?" *Proceedings of the VLDB Endowment*, vol. 16, no. 4, pp. 738–746, 2022.

[32] W. Ni, K. Zhang, X. Miao, X. Zhao, Y. Wu, Y. Wang, and J. Yin, "ZeroED: Hybrid zero-shot error detection through large language model reasoning," in *Proceedings of the IEEE 41st International Conference on Data Engineering (ICDE)*, 2025, pp. 3126–3139.

[33] S. Shankar, T. Chambers, T. Shah, A. G. Parameswaran, and E. Wu, "DocETL: Agentic query rewriting and evaluation for complex document processing," *Proceedings of the VLDB Endowment*, vol. 18, no. 9, pp. 3035–3048, 2025.

[34] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, "ReAct: Synergizing reasoning and acting in language models," in *International Conference on Learning Representations*, 2023.

[35] L. Wang, C. Ma, X. Feng, Z. Zhang, H. Yang, J. Zhang, Z. Chen, J. Tang, X. Chen, Y. Lin, W. X. Zhao, Z. Wei, and J.-R. Wen, "A survey on large language model based autonomous agents," *Frontiers of Computer Science*, vol. 18, no. 6, p. 186345, 2024.

[36] P. Li, X. Rao, J. Blase, Y. Zhang, X. Chu, and C. Zhang, "CleanML: A study for evaluating the impact of data cleaning on ML classification tasks," in *Proceedings of the IEEE 37th International Conference on Data Engineering (ICDE)*, 2021, pp. 13–24.

[37] S. Guha, F. A. Khan, J. Stoyanovich, and S. Schelter, "Automated data cleaning can hurt fairness in machine learning-based decision making," in *Proceedings of the IEEE 39th International Conference on Data Engineering (ICDE)*, 2023, pp. 3747–3754.

[38] D. Lopez-Paz and M. Oquab, "Revisiting classifier two-sample tests," in *International Conference on Learning Representations*, 2017.

[39] R. Shokri, M. Stronati, C. Song, and V. Shmatikov, "Membership inference attacks against machine learning models," in *2017 IEEE Symposium on Security and Privacy*, 2017, pp. 3–18.

[40] A. Machanavajjhala, J. Gehrke, D. Kifer, and M. Venkitasubramaniam, "ℓ-diversity: Privacy beyond k-anonymity," in *Proceedings of the 22nd International Conference on Data Engineering (ICDE)*, 2006, p. 24.

[41] N. Li, T. Li, and S. Venkatasubramanian, "t-closeness: Privacy beyond k-anonymity and ℓ-diversity," in *Proceedings of the IEEE 23rd International Conference on Data Engineering (ICDE)*, 2007, pp. 106–115.

[42] C. Dwork and A. Roth, *The Algorithmic Foundations of Differential Privacy*, ser. Foundations and Trends in Theoretical Computer Science. Now Publishers, 2014, vol. 9, no. 3–4.

[43] M. J. Sheller, B. Edwards, G. A. Reina, J. Martin, S. Pati, A. Kotrotsou, M. Milchenko, W. Xu, D. Marcus, R. R. Colen, and S. Bakas, "Federated learning in medicine: Facilitating multi-institutional collaborations without sharing patient data," *Scientific Reports*, vol. 10, p. 12598, 2020.

[44] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Ag"uera y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 54. PMLR, 2017, pp. 1273–1282. [Online]. Available: https://proceedings.mlr.press/v54/mcmahan17a.html

[45] P. Kairouz, H. B. McMahan, B. Avent *et al.*, "Advances and open problems in federated learning," *Foundations and Trends in Machine Learning*, vol. 14, no. 1–2, pp. 1–210, 2021.

[46] K. Bonawitz, V. Ivanov, B. Kreuter, A. Marcedone, H. B. McMahan, S. Patel, D. Ramage, A. Segal, and K. Seth, "Practical secure aggregation for privacy-preserving machine learning," in *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security*, 2017, pp. 1175–1191.

[47] Q. Chen, Y. Chen, B. Yan, X. Jiang, X. Zhang, Y. Kang, T. Zhang, W. Huang, C. Gao, L. Fan, and Q. Yang, "Model trip: Enhancing privacy and fairness in model fusion across multi-federations for trustworthy global healthcare," in *Proceedings of the IEEE 40th International Conference on Data Engineering (ICDE)*, 2024, pp. 2231–2244.

[48] A. Janosi, W. Steinbrunn, M. Pfisterer, and R. Detrano, "Heart disease," UCI Machine Learning Repository, 1989.

[49] W. Wolberg, O. Mangasarian, N. Street, and W. Street, "Breast cancer wisconsin (diagnostic)," UCI Machine Learning Repository, 1993.

[50] "Heart failure clinical records," UCI Machine Learning Repository, 2020.
[51] B. Ramana and N. Venkateswarlu, "ILPD (indian liver patient dataset)," UCI Machine Learning Repository, 2022.
[52] M. Little, "Parkinsons," UCI Machine Learning Repository, 2007.
[53] J. W. Smith, J. E. Everhart, W. C. Dickson, W. C. Knowler, and R. S. Johannes, "Using the ADAP learning algorithm to forecast the onset of diabetes mellitus," in *Proceedings of the Annual Symposium on Computer Application in Medical Care*, 1988, pp. 261–265.
[54] M. Patrício, J. Pereira, J. Crisóstomo, P. Matafome, M. Gomes, R. Seiça, and F. Caramelo, "Using resistin, glucose, age and BMI to predict the presence of breast cancer," *BMC Cancer*, vol. 18, p. 29, 2018.
[55] "Hepatitis," UCI Machine Learning Repository, 1983.
[56] L. Rubini, P. Soundarapandian, and P. Eswaran, "Chronic kidney disease," UCI Machine Learning Repository, 2015.
[57] M. Zieba, J. M. Tomczak, M. Lubicz, and J. Świątek, "Boosted SVM for extracting rules from imbalanced data in application to prediction of the post-operative life expectancy in the lung cancer patients," *Applied Soft Computing*, vol. 14, pp. 99–108, 2014.
[58] N. Seedat, J. Crabbé, I. Bica, and M. van der Schaar, "Data-IQ: Characterizing subgroups with heterogeneous outcomes in tabular data," in *Advances in Neural Information Processing Systems*, vol. 35, 2022.
[59] H. He, Y. Bai, E. A. Garcia, and S. Li, "ADASYN: Adaptive synthetic sampling approach for imbalanced learning," in *Proceedings of the IEEE International Joint Conference on Neural Networks (IJCNN)*, 2008, pp. 1322–1328.
[60] H. Han, W.-Y. Wang, and B.-H. Mao, "Borderline-SMOTE: A new over-sampling method in imbalanced data sets learning," in *Advances in Intelligent Computing (ICIC)*, ser. Lecture Notes in Computer Science, vol. 3644, 2005, pp. 878–887.
[61] N. Patki, R. Wedge, and K. Veeramachaneni, "The synthetic data vault," in *Proceedings of the IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, 2016, pp. 399–410.
[62] B. Nowok, G. M. Raab, and C. Dibben, "synthpop: Bespoke creation of synthetic data in R," *Journal of Statistical Software*, vol. 74, no. 11, pp. 1–26, 2016.
[63] D. S. Watson, K. Blesch, J. Kapar, and M. N. Wright, "Adversarial random forests for density estimation and generative modeling," in *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics (AISTATS)*, ser. Proceedings of Machine Learning Research, vol. 206, 2023, pp. 5357–5375.
[64] J. Demšar, "Statistical comparisons of classifiers over multiple data sets," *Journal of Machine Learning Research*, vol. 7, pp. 1–30, 2006.
[65] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.